

COSMO® Cube

Hardware Reference Manual — v1.1 — Engineering / Internal — LOT Systems, Inc.

1. Overview

The COSMO® Cube is a personal ambient intelligence device connected to lot-systems.com. It is a hardware API consumer: it polls GET /api/hardware/notifications and posts to POST /api/hardware/log. Full specification: COSMO-DEVICE-SPEC-v1.md. Full BOM with supplier links: COSMO-BOM-v1.md. Firmware: COSMO-FIRMWARE-v1.md. API: COSMO-SOFTWARE-API-v1.md. Manufacturing: COSMO-MANUFACTURING-v1.md. Charger: COSMO-CHARGER-SPEC-v1.md.

2. Physical Specification

Parameter	Value
Dimensions	40mm x 40mm x 5mm
Weight	~28g
Body material	316L Stainless Steel, CNC machined, 2 parts
Side A (back)	Mirror-polished #8 finish, LOT(R) engraved
Side B (front)	Satin finish - display, camera, button
Water resistance	IP54
Operating temp	0C to +45C

3. Electronics

Subsystem	Part	Interface
MCU	ESP32-S3-MINI-1U (dual-core, WiFi+BLE5)	-
Display	SSD1327 OLED 1.0" 128x128, 16-gray	SPI
Camera	Himax HM01B0, 320x320, 1.1mW	DVP
Weather	Bosch BME280 (temp/humidity/pressure)	I2C
IMU (AI-grade)	TDK ICM-42688-P, 6-axis	SPI
Light + gesture	Broadcom APDS-9960	I2C
Wireless charge Rx	TI BQ51013B, Qi 5W, 30mm coil	-
PMIC	TI BQ25892	I2C
Battery	Custom LiPo 280mAh, 35x35x2.5mm	JST-PH

4. LOT Platform Integration

The Copy button captures a sensor snapshot (temperature, humidity, pressure, light, orientation), optionally a camera frame, and POSTs it to `lot-systems.com/api/hardware/log`. The entry appears in the Log tab tagged [COSMO® Cube]. The device polls `lot-systems.com/api/hardware/notifications` every 60 seconds; the LOT AI (QI-46 Engine) pushes short messages such as “Coffee time!” that render on the OLED for up to 30 seconds. All calls use TLS 1.3 with a pinned certificate and a per-device API key stored in encrypted NVS.

5. Power Budget

Condition	Current	Life
Active WiFi, display on	~180mA	~1.5h
WiFi poll 60s, display on	~45mA	~6h
Deep sleep, wake every 60s	~2mA	~5.8 days
Charging (Qi 5W)	-	~2.5h full

6. Bill of Materials Summary (100-unit run)

Category	Est. 100-unit total
Electronics (MCU, display, camera, sensors, power)	~\$3,565
PCB fabrication + SMT assembly (PCBWay)	~\$1,650
SS enclosure, 2-part CNC (PCBWay)	~\$3,800
Wireless charger (Tx pad)	~\$900
Packaging	~\$400
Subtotal	~\$10,750
Contingency (15%)	~\$1,613
Grand total	~\$12,363
Target retail price	\$349/unit

Full line-item BOM with manufacturer part numbers, suppliers, and buy links: `COSMO-BOM-v1.md` §1–12.

7. Manufacturing Path

PCBWay is the primary manufacturer for all three streams: 4-layer PCB fabrication + turnkey SMT assembly, and CNC machining of the 316L stainless steel back plate (mirror #8) and front bezel (satin, with display/camera/button/mesh apertures). Target build: 10-week schedule, 10-unit prototype before the 100-unit run. Full QA checklist (14 items per unit): `COSMO-MANUFACTURING-v1.md` §3.3.

8. Roadmap

Phase	Status
Phase 0 - Design	Complete (this document set)

Phase 1 - Engineering (schematic, PCB layout, CAD, backend routes)	Not started
Phase 2 - Prototype (10 units)	Not started
Phase 3 - Production (100 units)	Not started
Phase 4 - Launch (My Devices page, OTA, FCC/CE, retail)	Not started

*COSMO® CIA — LOT Systems, Inc. — Inventor: Vadim Marmeladov — Originally specified 2026-06-12, landed on master 2026-09-16.
See docs/benchmark/LOT-SR-20260916-01.md for the session record.*